Information, CRLB, sensitivity and bandwidth for the NV resonance-search experiment

NVision

1 The experiment

A microwave drive of frequency x is applied and the NV fluorescence is read out. The signal has a dip centred on an unknown parameter θ (the `zeeman_shift`, in Hz) whose value is set by the magnetic field B. Each *step* is one measurement at a chosen x_t, with k_t averaged shots and Gaussian noise of standard deviation σ per shot. The goal is a small uncertainty $\delta\theta$, hence δB, in as little measurement time as possible. Throughout, $g(x) = \partial\,\mathrm{signal}(x;\theta)/\partial\theta$ is the slope of the signal with respect to θ.

2 Fisher information of one measurement

For Gaussian noise a measurement at x carries

$$I(x) = \frac{k\,g(x)^2}{\sigma^2} \qquad [\mathrm{Hz}^{-2}]. \tag{1}$$

It is largest where the signal changes fastest with θ, the steepest flank of the dip, and zero at the dip centre and far in the wings. For a Lorentzian dip of half-width γ and depth C the maximum is at $|x-\theta| = \gamma/\sqrt{3}$ with $|g_{\max}| = \frac{3\sqrt{3}}{8}\,C/\gamma \approx 0.65\,C/\gamma$, so, with $\mathrm{SNR} = C/\sigma$ and one shot,

$$I^* = \frac{g_{\max}^2}{\sigma^2} \approx 0.42\,\frac{\mathrm{SNR}^2}{\gamma^2}. \tag{2}$$

The per-step *efficiency* $\eta_t = g(x_t)^2/g_{\max}^2 \in [0,1]$ is the share of the best possible information a measurement collects.

3 The Cramér–Rao bound

Independent measurements add information, so n measurements give $J_n = \sum_t I(x_t) \le n\,I^*$, and for an unbiased estimator

$$\delta\theta \;\ge\; \frac{1}{\sqrt{J_n}} \;\ge\; \frac{1}{\sqrt{n\,I^*}}. \tag{3}$$

The right-hand side is the bound used in NVision: the oracle that always measures at the best point of the true signal. It is a floor for any locator, including adaptive ones (the information of an adaptive design is still capped by I^* per step). On log–log axes it is a straight line of slope $-1/2$. A locator is judged by its *Fisher efficiency* (bound steps over steps spent), which is 1 only if every measurement sits at the best point.

4 The prior is information

The bound above is for the likelihood alone. A Bayesian posterior also starts from a prior, and a search window is a prior: a uniform window of width W has variance $W^2/12$. The right bound is the van Trees form

$$\delta\theta(n) \;\geq\; \left(J_0 + n\,I^*\right)^{-1/2}, \qquad J_0 = \sigma_0^{-2}, \tag{4}$$

with σ_0 the prior standard deviation ($W/\sqrt{12}$ for a uniform window, a variance-matched convention, not the exact Fisher information of the prior). It is flat while the prior dominates and approaches the $-1/2$ line from *below*. So a claimed $\delta\theta$ under $1/\sqrt{nI^*}$ at small n is expected, and a narrower window starts lower for free. Only the true error averaged over repeats falling below the van Trees line would indicate a defect. A focus window built from the search stage is a prior that the search paid for; the cumulative clock and information must therefore include the search shots.

5 Stages

Three regimes follow from this. **Search**: nothing is known about where the dip is, and the Fisher bound does not apply. **Focus**: the information per shot rises towards I^* as the posterior narrows, so $\delta\theta$ falls *faster* than $n^{-1/2}$ while still above the bound. **Refine**: every measurement is near the best point, $J_n \approx J_0 + nI^*$, and $\delta\theta \propto n^{-1/2}$. The oracle step counts, with ε the half-width around the best point where $\eta \geq 0.9$ and τ the stop target, are

$$N_{\text{focus}} \geq \frac{\varepsilon^{-2} - J_0}{I^*}, \qquad N_{\text{refine}} \geq \frac{\tau^{-2} - \varepsilon^{-2}}{I^*}, \qquad N_{\text{total}} \geq \frac{\tau^{-2} - J_0}{I^*}. \tag{5}$$

6 Sensitivity

Let t_s be the time per step including overhead, so $T = n\,t_s$. In the asymptotic regime $\delta\theta = 1/\sqrt{nI^*} = \sqrt{t_s/(I^*T)}$, so

$$\eta_\theta \;=\; \delta\theta\,\sqrt{T} \;=\; \sqrt{\frac{t_s}{I^*}} \qquad [\text{Hz}\,\sqrt{\text{s}}]. \tag{6}$$

The magnetic sensitivity is $\eta_B = \eta_\theta/\kappa$ in $\text{T}/\sqrt{\text{Hz}}$, where $\theta = \kappa B$ (for a single-dip shift $\kappa = \gamma_e \approx 28.0\,\text{GHz/T}$; the factor depends on how `zeeman_shift` is defined and must be set explicitly). η is a constant of the locator only when $\delta\theta \propto T^{-1/2}$ holds exactly, that is, in refine. Before that, $\delta\theta\sqrt{T}$ is not constant, and its instantaneous form

$$\eta_{\text{inst}}(t) = \left(\frac{\mathrm{d}J}{\mathrm{d}t}\right)^{-1/2}, \qquad J = \delta\theta^{-2}, \tag{7}$$

falls during focus towards η_θ. The ratio $\eta_\theta^2/\eta_{\text{inst}}^2 \leq 1$ is the Fisher efficiency.

7 Bandwidth

Bandwidth is the rate at which the field can be re-measured to a required precision δB^*. The operational definition is the first-passage time $T(\delta B^*)$ at which the *true* error first stays below the

target, and

$$f_{\mathrm{BW}} = \frac{1}{T(\delta B^*)}. \tag{8}$$

It needs no assumption about the scaling and covers search, focus and refine together. In refine, with constant η_B, it reduces to

$$T(\delta B^*) = \left(\frac{\eta_B}{\delta B^*}\right)^2, \qquad f_{\mathrm{BW}} = \left(\frac{\delta B^*}{\eta_B}\right)^2. \tag{9}$$

A convention (for example $1/(2T)$) can be chosen later, but should be stated with every number. The effective sensitivity at a target, $\eta_{\mathrm{eff}}(\delta B^*) = \delta B^* \sqrt{T(\delta B^*)}$, tends to η_B as $\delta B^* \to 0$ and is not a constant otherwise. NVision stores $T(\tau)$ in steps (`stage_true_passage_step`) and $\tau\sqrt{T}$ (`stage_eta_eff`).

8 Summary

Quantity	Definition	Holds when
Information per step	$I = k\,g^2/\sigma^2$, best I^*	always
CRLB (oracle)	$\delta\theta \geq (nI^*)^{-1/2}$	likelihood only, unbiased
Van Trees	$\delta\theta \geq (J_0 + nI^*)^{-1/2}$	Bayesian posterior with a prior
Sensitivity	$\eta_\theta = \sqrt{t_s/I^*}$	refine (J linear in t)
Instantaneous sensitivity	$\eta_{\mathrm{inst}} = (\mathrm{d}J/\mathrm{d}t)^{-1/2}$	any stage
Bandwidth	$1/T(\delta B^*)$, first passage of true error	any stage